

Optimal Ad Impression Allocation under Fine Targeting, Campaign Quotas, and Finite Supply

When narrow targets contend for scarce inventory, per-event greedy serving forfeits revenue; an offline allocation plan, executed online in constant time, recovers it.

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ABSTRACT

A mobile network operator can serve advertising far more targeted and fine-grained than any web platform: subscription data supplies verified demographics and financial signals, and gateway-observed usage supplies complete browsing histories and click attribution under a single stable identity per subscriber. Advertisers exploit that precision by buying narrow, high-value targets. The setting imposes hard constraints: each campaign carries a contracted impression quota that must be delivered within its flight; targeting restricts every campaign to specific audiences, time windows, and content categories; per-subscriber frequency caps bound exposures across all inventory; subscribers arrive in unpredictable order; and each decision must complete within the latency of a page load. Precision concentrates demand: finely targeted campaigns compete for the same scarce sub-audiences and hours, so every impression granted to one campaign is a measurable loss to another. The greedy rule of serving whichever ad the viewer is most likely to click therefore systematically misallocates this contested inventory. Fine targeting thus creates the allocation problem, while complete attribution makes it solvable: click probabilities per campaign and segment are measured, not guessed, so the trade-off between any two assignments is a number. This report presents a two-stage architecture that exploits both facts. An offline stage clusters subscribers by profile and response behavior, estimates click probabilities per campaign and segment, forecasts traffic, and solves a linear program that

allocates every campaign's quota across segments for maximum expected revenue. The resulting plan is compiled into bit-masked lookup tables that an online stage executes per impression in constant time, blending cluster-level click rates through soft-membership weights and steering delivery back to the planned trajectory with a performance index. On a minimal worked instance, planned allocation raises expected clicks 11% over greedy serving with identical inputs. In Monte-Carlo simulation at realistic scale (25 campaigns, 60 segments, 200,000 impressions per run, 20 replications), the planned policy delivers a mean 6.5% revenue lift and 33% more expected clicks than greedy serving on identical event streams, and captures 94% of the offline optimum computed with hindsight, despite serving against forecast rather than realized traffic. On real logs (26.6M impressions of the public Taobao display-advertising dataset), walk-forward replay over five independent serving days confirms the architecture end-to-end: the plan beats greedy serving by 3.5% expected clicks ($z = 5.0$, every fold positive) and beats an adaptive dual-price pacing baseline by 0.9% ($z = 2.0$), on a configuration fixed in advance by a measurable design rule; under over-subscription stress the plan's advantage over adaptive pacing grows to 5–7%, and periodic re-solving of the same plan, the architecture's pacing loop made quantitative, is the best policy tested. The rule behind the configuration: segment on the axes whose campaign-by-segment interaction persists across planning windows. Served responses feed the next planning cycle, closing the loop.

1 Introduction

A mobile advertising platform must answer one question millions of times a day: *a subscriber has just opened a page; which campaign should fill this impression?* The obvious answer, show whichever ad this subscriber is most likely to click, is wrong. Campaigns carry contracted impression quotas, subscribers arrive in an unpredictable order, and every impression given to one campaign is an impression denied to all others. Serving each impression greedily fills popular campaigns with the first subscribers who arrive, not those who respond best, and leaves the remaining campaigns with a poorly matched audience.

On the open web this failure is blunted by coarseness. A cookie-based ad network targets broad, loosely defined audiences, because that is all its data supports: identity is a resettable browser artifact, demographics are inferred, and each network attributes responses only on the slice of the web it instruments. Broad audiences are interchangeable audiences: most impressions have several roughly equivalent uses, so the opportunity cost of a greedy choice

is small. Operator-run mobile advertising inverts every one of these conditions. The operator holds verified declared demographics, financial and purchase signals, network-observed location, and, because all browsing crosses its gateway, **complete response attribution**: every impression and every click, on every site, joined under one stable subscriber identity. Data of this quality supports far finer segmentation, and finer segmentation is precisely what advertisers pay an operator for: not “males 18–49” but urban music buyers on sports content on weekend afternoons.

Precision is what turns serving into an allocation problem. Narrow, high-value targets are scarce, and campaigns defined this precisely overlap: the subscribers one advertiser most wants are the subscribers other advertisers most want, in the same hours, on the same content. Contention over specific sub-audiences stops being an edge case and becomes the normal operating condition. Every impression then carries a real opportunity cost, and greedy serving, harmless when audiences are interchangeable, systematically hands the scarcest inventory to whichever campaign arrives first. The same richness that creates the problem also makes it solvable: with complete attribution, the click probability of every (campaign, segment) pair is a measured quantity, so the cost of any assignment against any alternative can be computed rather than guessed. An operator that targets precisely and attributes completely does not merely benefit from global optimization; it forfeits the value of its own data without it. Precision here is a property of *demand*: advertisers buy narrow targets, and narrow targets contend. Which segmentation the allocator itself should use is a separate, empirical question, and Section 8 shows its answer is governed by which interactions persist, not by maximal fineness.

This report describes an architecture that treats ad serving as that **global allocation problem**. An offline analytics stage learns who responds to what, clusters subscribers by profile and response behavior, predicts traffic per segment, and solves for an allocation plan that maximizes expected revenue across all campaigns at once. The plan is compiled into compact lookup tables. An online decision stage then serves each impression in constant time by consulting those tables, tracking actual delivery against plan, and steering each campaign back toward its planned trajectory. Feedback from served impressions flows back to analytics, closing the loop and refining the model each cycle. Section 2 details the operator’s data advantage, Section 3 quantifies the cost of greedy serving on a minimal instance, Sections 4–6 develop the formulation and both stages, Section 7 confirms the revenue case by simulation, Section 8 validates it by replaying 26.6 million impressions of a

public advertising log, and Section 9 situates the design in the online-allocation and offline-evaluation literature.

2 The operator’s vantage point

Mobile advertising delivered by the network operator itself is a different discipline from advertising on the open web, and the difference is the data. A web ad network observes a browser: an unverified, resettable cookie, activity on the sites it happens to instrument, and demographics inferred by guesswork. A network operator observes a *customer*. The subscriber relationship supplies a verified identity anchored in a billing account, declared attributes such as age, gender, and home address, and financial signals such as plan type, spending level, and purchase history for content and app spending. The network itself supplies what no publisher-side system can assemble: every browsing session crosses the operator’s gateway regardless of destination site, so the response history is complete rather than sampled from one publisher’s corner of the web, and it is joined under one stable identifier rather than stitched probabilistically across cookies and devices.

Dimension	Web ad network	Network operator
Identity	Cookie or device fingerprint; resettable, unverified	Subscriber account; stable, verified, one per person
Demographics	Inferred from behavior	Declared at subscription: age, gender, address
Financial signal	None or modeled	Plan, spend level, content purchase history
Location	Coarse, via IP	Network-observed, continuous
Browsing coverage	Sites the network instruments	All traffic through the gateway
Exposure control	Per-site frequency caps	Global caps across all inventory, per subscriber

This vantage point changes what is *possible*, and each advantage maps directly onto a component of the architecture in this report. Verified profiles and complete response

histories make the proximity metric of Section 5.2 meaningful: subscribers are clustered on real attributes and full behavioral records, not on noisy proxies, so the clusters are stable enough to pool click statistics over. Complete traffic visibility makes the context slots and traffic forecasts of Section 5.3 accurate: the operator sees the whole denominator, every impression opportunity, not just the ones flowing through partnered sites. The single stable identifier makes global exposure policy enforceable: a frequency cap means the subscriber saw at most so many ads today *anywhere*, which no cookie-scoped system can promise. And the billing relationship closes the loop commercially, since the same account that receives the ad can redeem the offer.

Two consequences deserve emphasis, because the allocation machinery depends on them quantitatively. First, **segmentation can be far finer**. Cluster granularity is limited by statistics: a segment must generate enough impressions and clicks for its estimated rates to be trustworthy. Because the operator's data is complete (every subscriber, every session) and clean (verified attributes, deduplicated identity), the same statistical reliability is reached with much smaller segments than inferred web data allows. Finer segments mean click probabilities that track real behavioral differences instead of averaging over them, and every refinement of $p_{c,s}$ translates directly into allocation-plan revenue. Second, **overall load is predictable**. The operator observes total traffic, its daily and weekly rhythm, per content category and per region, so the supply forecast H_s is estimated from a census rather than a sample. Accurate supply forecasts are what make an offline plan trustworthy at all: a plan allocating impressions that never materialize, or missing volume that does, must be repaired constantly at serving time. Rich subscriber knowledge sharpens both sides of the optimization, the value of each pairing and the volume available to pair.

The vantage point also carries obligations that shape the design. This data is usable for advertising only within a consent and data-protection framework: profiling subscribers requires a lawful basis and per-subscriber opt-out under regimes such as the GDPR and ePrivacy rules, and gateway-observed browsing is subject to purpose limitation. The architecture is built for data minimization under those constraints. Raw attributes and browsing records are confined to the offline stage; the serving path receives only anonymous cluster distances and campaign eligibility lists, and every online decision reads aggregate tables rather than individual histories. Segment coarseness is itself a privacy parameter: the segments that carry allocation value in Section 8 are a handful of aggregate behavior tiers, not individual dossiers. Not every subscriber has a profile at all: prepaid

subscribers may be anonymous, which is why the clustering machinery supports membership computed from response behavior alone (Section 5.3).

Rich knowledge, however, only determines how well each individual impression *could* perform. It says nothing about how impressions should be divided among competing campaigns. That is a separate problem, and it is where the revenue argument begins.

3 Motivation: the cost of deciding one impression at a time

Consider the smallest interesting instance. Two campaigns run at the same time: a soft drink and a sports car. Each has bought exactly **two impressions**. Four subscribers will each generate one ad opportunity. From past behavior we can estimate every subscriber's probability of clicking each campaign:

Subscriber	P(click drink)	P(click car)
u_1	0.10	0.80
u_2	0.30	0.50
u_3	0.20	0.60
u_4	0.40	0.70

A greedy server shows every subscriber their personally best ad. All four prefer the car, so the car campaign's quota is consumed by whoever happens to arrive first, say u_1 and u_2 . The drink campaign is then forced onto the remaining two:

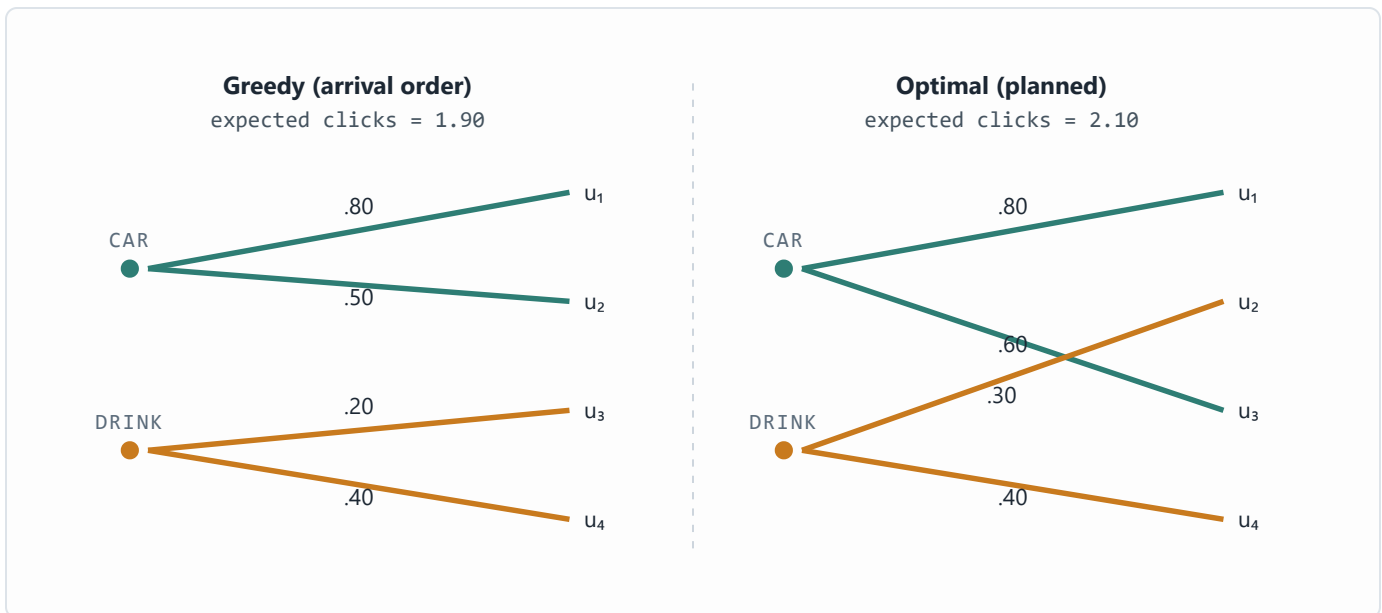


Figure 1. Two campaigns, four subscribers, two impressions per campaign. Greedy serving fills the car campaign with the first arrivals (u_1, u_2) for $0.80 + 0.50 + 0.20 + 0.40 = 1.90$ expected clicks. The optimal assignment gives the car campaign u_1 and u_3 and the drink campaign u_2 and u_4 , for $0.80 + 0.60 + 0.30 + 0.40 = 2.10$, an 11% lift from re-pairing the same impressions. The greedy value depends on arrival order; averaged over all orders it is 1.80, so the drawn order is a conservative case.

Nothing about the subscribers or the campaigns changed between the two panels. The only difference is that the right-hand assignment was chosen *jointly*: the car campaign gives up u_2 (a 0.50 click chance), whose 0.30 with the drink remains a strong pairing, and takes u_3 (0.60) instead. Each individual swap looks locally worse for someone; the total is strictly better.

The gap grows with scale. With hundreds of campaigns, millions of subscribers, targeting constraints, frequency caps, and time-of-day windows, greedy serving does not just lose a few percent: it systematically starves narrow-audience campaigns, over-exposes indifferent subscribers, and leaves contracted quotas unfilled at the end of the flight. Optimal allocation is not a refinement of ad serving; it is the difference between selling inventory and wasting it.

The core insight: click probability is a property of a *pair* (audience, campaign), not of a campaign alone. Revenue is maximized by choosing the pairing globally, subject to each campaign's quota, and this choice can be made ahead of time because audience behavior is statistically predictable even though individual events are not.

4 Problem formulation

Formally, the planning stage solves a transportation-style optimization. Audience supply is described by segments: s ranges over (cluster, context slot) pairs, where a cluster is a group of behaviorally similar subscribers and a context slot is a recurring situation (a time window crossed with a content category). Let H_s be the predicted number of impressions segment s will generate during the planning horizon, and $p_{c,s}$ the estimated probability that an impression from segment s produces a click on campaign c . Each campaign pays r_c per impression and q_c per click, and has bought Q_c impressions. The plan chooses $x_{c,s}$, the number of impressions of campaign c allocated to segment s :

$$\begin{aligned}
 &\text{maximize} && \sum_{c,s} x_{c,s} \cdot (r_c + q_c \cdot p_{c,s}) \\
 &\text{subject to} && \sum_c x_{c,s} \leq H_s && \text{(segment supply)} \\
 &&& \sum_s x_{c,s} \leq Q_c && \text{(campaign quota)} \\
 &&& x_{c,s} = 0 \text{ where targeting excludes } (c, s) \\
 &&& x_{c,s} \geq 0
 \end{aligned}$$

Figure 1 is this program with two campaigns, four single-subscriber segments, $H_s = 1$ and $Q_c = 2$. At production scale the same structure holds; the segments are just coarser and the counts larger. Because both objective and constraints are linear, the plan is solvable with standard simplex or interior-point solvers [1] even for large instances, and, critically, it is solved **offline**. The online server never optimizes; it executes a pre-computed plan and corrects for drift.

Three practical notes. First, the objective prices impressions and clicks separately, so campaigns that pay mostly per impression and campaigns that pay mostly per click compete in one currency, expected revenue; when supply suffices, quotas bind at the optimum (every unit of x earns at least $r_c > 0$), and the impression-price term becomes a constant, making the allocation choice a pure expected-clicks problem. Second, the quota appears as an upper bound: when targeting and supply cannot fill a quota, the program returns the revenue-maximal under-delivery rather than infeasibility, and the shortfall is the input to make-good scheduling, which we treat as outside the model. Third, the quota constraint is what makes greedy serving inadequate: without it, serving every impression to its highest-value campaign would be optimal, and no planning stage would be needed.

5 The offline stage: learning the inputs to the plan

The allocation program needs two inputs that do not exist in raw data: predicted traffic per segment (H_s) and click probabilities per campaign-segment pair ($p_{c,s}$). The offline analytics stage builds both from event history, under three modeling assumptions, each holding with high probability rather than certainty:

1. The distribution of (context, profile) pairs observed in the past will recur in the future.
2. Subscribers who responded similarly under similar conditions will continue to respond similarly.
3. Subscribers who respond similarly share something observable in their profiles or histories.

Assumption 1 justifies forecasting traffic from history; assumptions 2 and 3 justify pooling sparse per-subscriber click data into clusters that have enough volume to estimate probabilities reliably.

5.1 Response records

The atomic unit of evidence is an ad response record: who saw what, in what context, and whether they clicked.

Time of day	Day of week	Content category	Campaign	Click
11:23	Monday	News	C1 (active)	0
14:15	Monday	Sport	C2 (active)	1
18:14	Sunday	Sport	C4 (past)	1

Records are grouped per subscriber and joined with the profile (age, gender, location, income band, purchase indicators) where one is available. Past campaigns matter as much as active ones: a subscriber's reaction to a finished campaign is evidence about how they will react to similar future ones.

5.2 Proximity and clustering

Subscribers are clustered by a weighted distance that mixes profile similarity with response-pattern similarity: response to the same campaigns (active and past) and to similar campaigns. Each attribute contributes its own distance component:

Attribute	Distance component
Age	Absolute difference
Gender	Fixed penalty if different
Campaign affinity	By campaign category; fixed penalty if different
Day of week	Absolute difference, circular
Location (x, y)	Straight-line distance

Membership is deliberately **soft**. A subscriber is not filed into one cluster; they carry a pre-computed distance to each nearby cluster. A 35-year-old married streaming subscriber in a large city may sit at distance 0.8 from the “metropolitan professionals” cluster and 0.1 from “music lovers”; a teenage in-app purchaser in the same city sits close to “kids” and “music lovers” instead. These distances become interpolation weights at serving time (Section 6), so predictions blend the behavior of every cluster the subscriber resembles.

5.3 Cluster representation and click estimation

Each cluster is summarized by two compact structures. Its *context slots* describe when and where the cluster generates traffic, with predicted hit counts; this is the supply side H_S . Its *click statistics* record impressions and clicks per campaign per context slot, the raw material for $p_{c,s}$:

Context slot	Time of day	Day of week	Category	Predicted hits
CS1	10:00–12:00	Mon–Tue	News, Sport	3,000
CS2	11:00–17:00	Sun	Sport	4,000

Campaign	Context slot	Impressions	Clicks
C1	CS1	1,000	3

Campaign	Context slot	Impressions	Clicks
C1	CS2	2,000	2
C2	CS1	1,000	7

For a brand-new campaign with no history of its own, probabilities are borrowed from the most similar past or active campaigns in the same category, the same pooling idea applied across campaigns instead of across subscribers. Where the profile itself is missing (anonymous or prepaid subscribers), distance to clusters is computed from response history alone, and the subscriber inherits the response statistics of the cluster they behave like. The design degrades gracefully as data thins out: per-subscriber evidence backs off to cluster evidence, and per-campaign evidence backs off to category evidence.

5.4 Compiling the plan into lookup tables

Solving the allocation program yields planned impression counts per (campaign, cluster, context slot). Everything the online server needs is then flattened into three lookup structures, so that no clustering, no distance computation, and no optimization happens on the serving path:

Structure	Keyed by	Contents
Model table	cluster × campaign	Targeted time slots and content categories as bit masks; planned impressions; expected clicks
Match table	subscriber	Campaigns this subscriber is eligible for under targeting rules
Cluster table	subscriber	Distance to each nearby cluster (the soft-membership weights)

Time and content targeting are encoded as fixed-width bit masks, 168 weekly hour-slots and up to 256 content categories, so an eligibility test at serving time is a bitwise AND rather than a rule evaluation. A model table row reads, in full: *“cluster L1, campaign C1, time mask 0100..., content mask 1111..., 10,000 planned impressions, 450 expected clicks.”* That one row carries targeting, allocation, and the performance baseline the online stage will pace against.

6 The online stage: executing the plan in real time

At serving time an impression event arrives carrying a subscriber identifier and a context (time, day, content category of the page being viewed). The decision engine must return a single campaign within the latency budget of a page load. It consults the three model structures plus two live state tables: a **status table** mirroring the model table but accumulating *actual* impressions and clicks, and a **response table** holding each subscriber's recent exposure records.

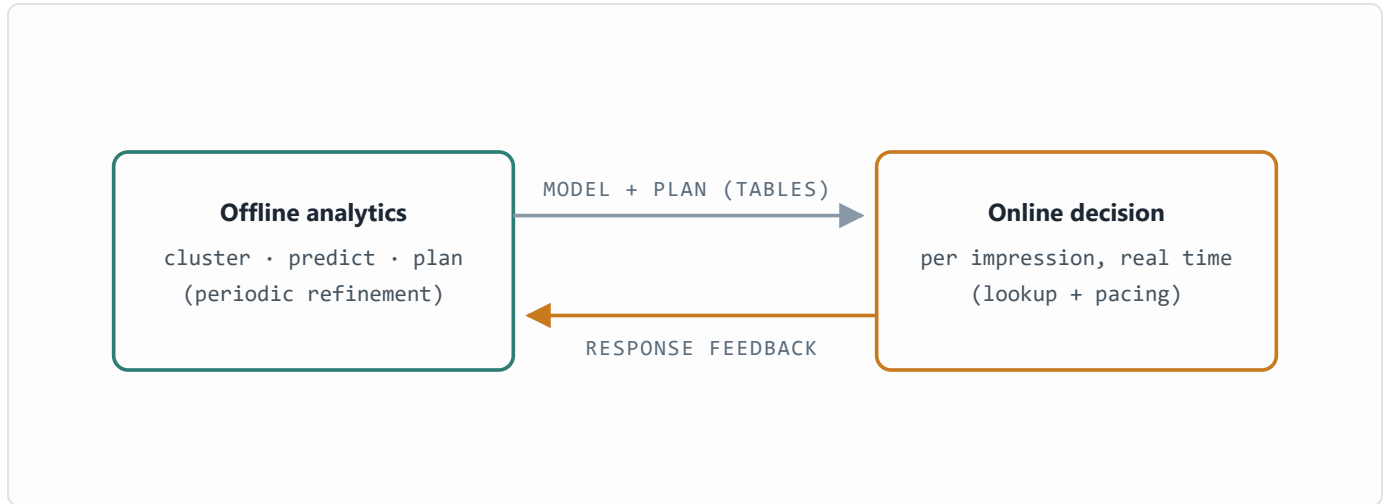


Figure 2. The closed loop. Offline analytics compiles the model and allocation plan into lookup tables; the online engine executes them per impression and streams response records back, which the next analytics cycle uses to refine clusters, probabilities, and the plan.

6.1 The seven-step decision

1 Find eligible campaigns

`match table` Look up the campaigns this subscriber may be shown under targeting rules.

2 Apply exposure policy

`response table` Drop campaigns that would violate frequency rules for this subscriber, a minimum interval between exposures, a maximum number of exposures per day.

3 Fetch cluster distances

`cluster table` Retrieve the subscriber's pre-computed distances to nearby clusters.

4 Predict click probability per candidate

`model table` Blend cluster-level click rates with the soft-membership weights (Section 6.2).

5 Compute pacing per candidate

`status table` Compare actual delivery against plan, prorated to this point in the flight (Section 6.3).

6 Decide

If any campaign is critically underserved, serve it. Otherwise drop overserved campaigns and serve the candidate with the greatest expected revenue.

7 Update state

`status + response tables` Increment the served campaign's actual counters and append the exposure record, which later feeds back to analytics.

Every step is a keyed lookup, a bit-mask test, or arithmetic over a handful of rows. The expensive work (clustering, probability estimation, optimization) was all paid for offline.

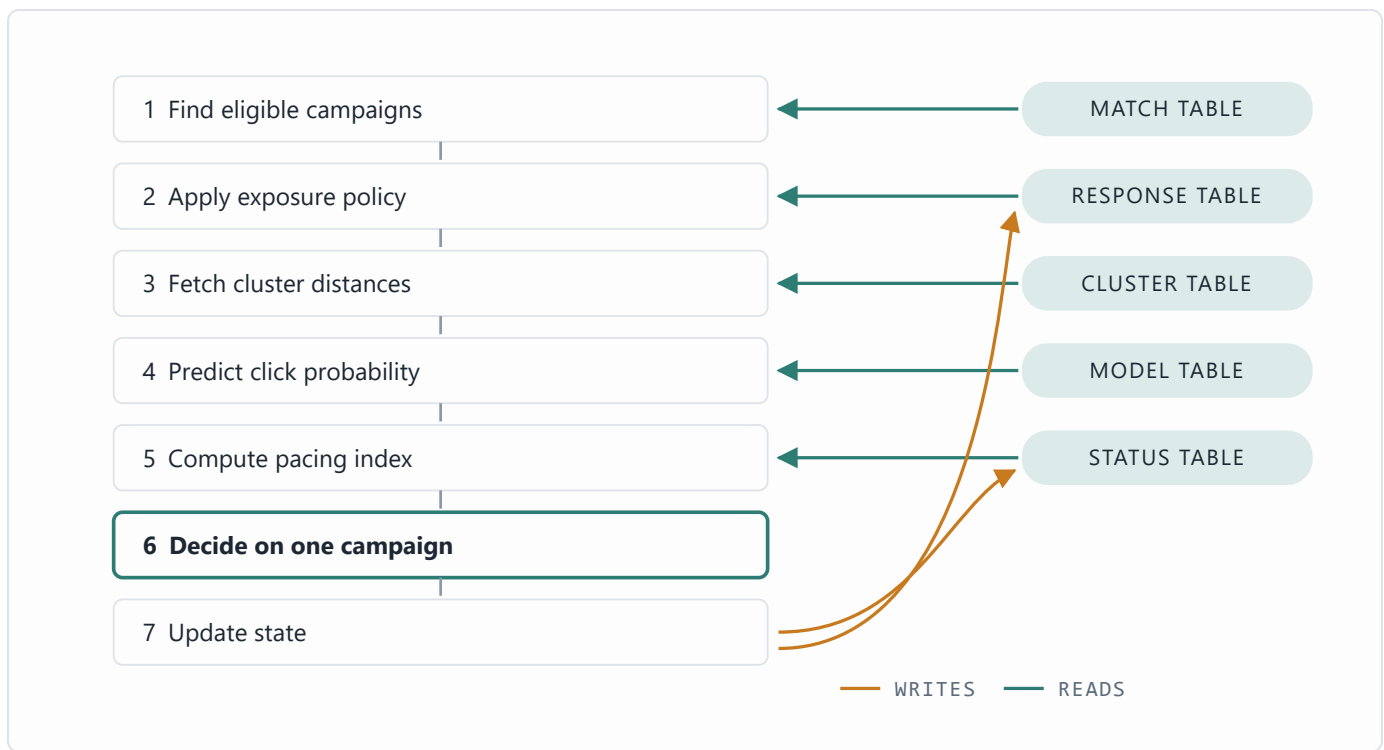


Figure 3. The seven-step decision path and its data dependencies. Steps 1–5 each read one pre-computed or live table; step 6 is pure arithmetic over the candidates; step 7 writes the served impression back into the status and response tables, which is also the record that feeds the next offline refinement cycle.

6.2 Click prediction by soft membership

Suppose the event context is *Monday, 14:00, Sport* and the subscriber sits at weight 0.2 to cluster L1 and 0.8 to cluster L2 (weights derived from the stored distances). The model table gives campaign C1’s historical rates in the matching context: 40 clicks per 8,000 impressions in L1 and 60 per 10,000 in L2. The prediction is the weighted blend:

$$p(\text{click} \mid C1) = 0.2 \cdot (40 / 8,000) + 0.8 \cdot (60 / 10,000) = 0.0058$$

Because the subscriber borrows from every cluster they resemble, the estimate is stable even for subscribers with thin personal histories, and it is computed from two table rows and four multiplications.

6.3 Pacing: planned versus actual

The plan is a trajectory, not just a total. If a campaign is one third of the way through its flight in a given cluster, it should have received roughly one third of its planned impressions there. The engine aggregates this across the clusters where the campaign runs:

Cluster	Flight elapsed	Planned by now	Actual
L1	33%	$8,000 \times 0.33 = 2,640$	2,000
L2	50%	$10,000 \times 0.50 = 5,000$	5,100
Total		7,640	7,100

$$\text{performance index} = \text{actual} / \text{planned-by-now} = 7,100 / 7,640 \approx 93\%$$

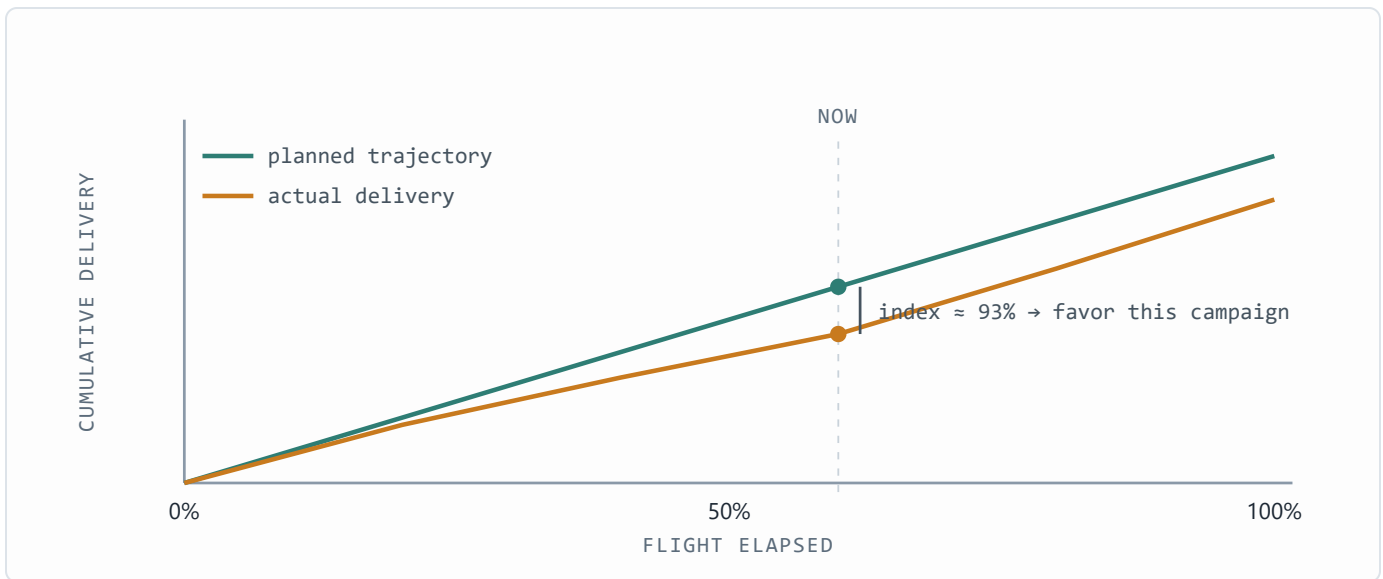


Figure 4. Pacing as trajectory tracking. The plan implies a cumulative delivery path; actual delivery drifts below it, the performance index measures the gap at the current moment, and the decision rule favors the lagging campaign on suitable impressions until the curves reconverge.

An index near 100% means the campaign is on plan. Far below a threshold (say 50%), the campaign is critically underserved and jumps the queue (step 6); far above another (say 120%), it is overserved and stands aside. Between the thresholds, campaigns compete on the same per-impression value the plan optimizes, expected revenue = impression price + click price × P(click):

Campaign	Impression price	Click price	Perf. index	P(click)	Expected revenue
C1	\$0.10	\$1.00	55%	0.01	\$0.110
C2	\$0.10	\$0.80	91%	0.03	\$0.124
C3	\$0.10	\$1.50	96%	0.02	\$0.130

Campaign	Impression price	Click price	Perf. index	P(click)	Expected revenue
C4	\$0.10	\$1.00	128%	0.07	\$0.170

C4 carries the greatest expected revenue but sits at 128% of plan, past the overserved threshold, so it is dropped; C1 at 55% is behind plan but above the critical threshold, so it earns no override. Among the remaining candidates the engine serves C3, the greatest expected revenue for this impression. The pacing mechanism is what lets a static offline plan survive contact with real, noisy traffic: the plan sets the destination, and the index steers thousands of small per-impression corrections toward it. Deviations that pacing cannot absorb (a traffic forecast that proves wrong, a campaign that outperforms its category prior) are exactly what the feedback loop carries back into the next planning cycle.

7 Simulation: planned allocation versus greedy serving

The four-subscriber example of Section 3 proves the mechanism exists; a Monte-Carlo simulation quantifies it at realistic scale. Each replication generates a synthetic market: 60 segments with lognormal traffic (about 200,000 impression opportunities), 25 campaigns whose quotas sell 80% of expected inventory, per-impression prices drawn uniformly on [\$0.05, \$0.30] and per-click prices on [\$0.50, \$3.00], and click probabilities from a campaign-audience affinity model (each campaign genuinely fits some audiences better than others, with rates spanning 0.1%–8%). Each campaign is eligible, by targeting, on roughly 60% of segments.

Two policies then serve the *identical* event stream under identical quotas, prices, probabilities, and targeting. The **greedy** policy is the conventional per-event rule: serve the eligible, quota-remaining campaign with the highest immediate expected revenue. The **planned** policy solves the linear program of Section 4 offline on the traffic forecast, then serves each event by drawing down the planned allocation, falling back to greedy where the plan is locally exhausted. The plan never sees the realized traffic; events are drawn randomly around the forecast, so the planned policy faces genuine forecast error. A third quantity, the **hindsight bound**, is the offline optimum computed on the traffic that actually occurred; no online policy can exceed it.

Measure (20 replications)	Mean	Range
Revenue lift, planned over greedy	+6.5%	+4.1% to +8.8%
Expected-click lift, planned over greedy	+33.0%	+22.9% to +40.6%
Planned revenue as share of hindsight bound	93.8%	$\geq 91.7\%$
Greedy revenue as share of hindsight bound	88.1%	$\geq 85.4\%$
Quota fill, both policies	100%	all replications

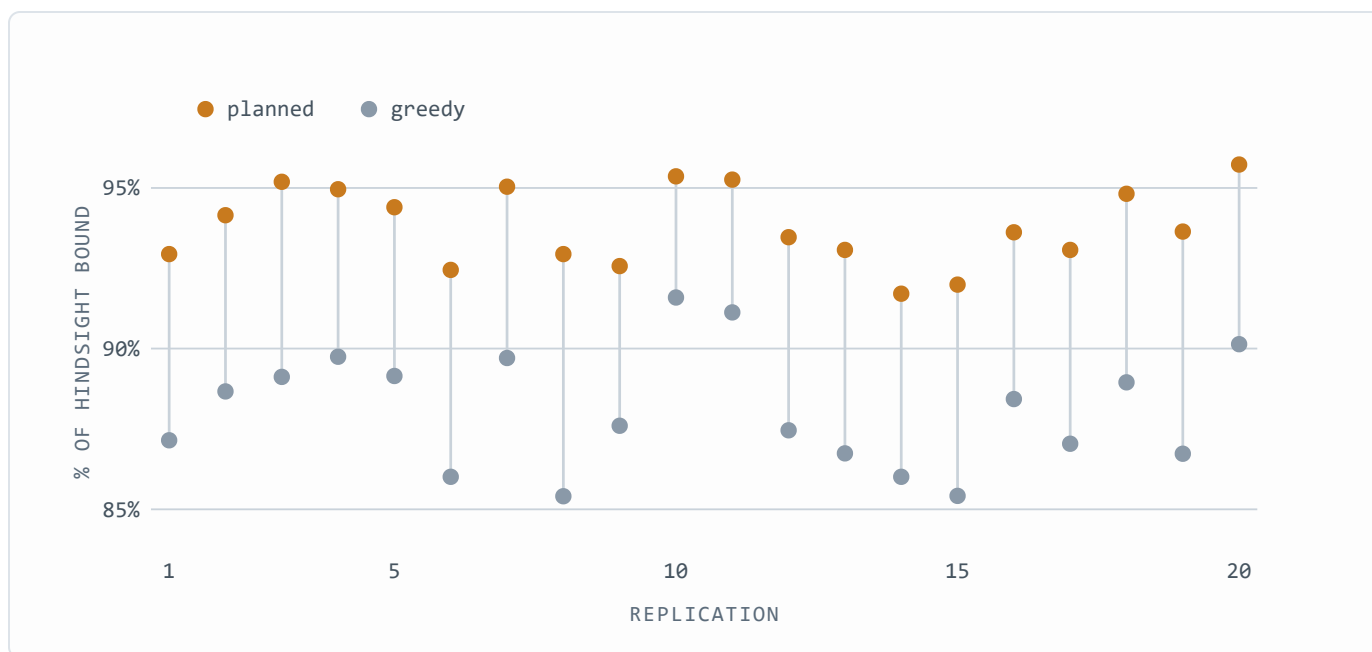


Figure 5. Revenue captured by each policy as a share of the hindsight offline optimum, per replication. Planned allocation dominates greedy serving in all 20 replications; the two distributions do not overlap (planned minimum 91.7%, greedy maximum 91.6%).

Both policies fill every quota, so the entire gap is *pairing quality*: the same impressions, sold to the same campaigns, assigned to different audiences. This also explains why the click lift (33%) exceeds the revenue lift (6.5%): per-impression fees are earned identically by both policies once quotas fill, so planning’s advantage is concentrated wholly in the click-priced component of revenue. In markets weighted further toward performance pricing, the revenue gap widens toward the click gap.

The simulation carries its own controls. Both policies are verified against the hindsight bound in every replication. On control instances where all campaigns share identical click probabilities, so pairing cannot matter, the measured lift is 0.000%, confirming that the gap

measures allocation quality and not an artifact of the harness. And the planned policy's 93.8% of a bound computed with perfect hindsight shows that forecast noise costs the plan only a few points, the residual that Section 6's pacing mechanism and the refinement loop exist to absorb.

8 Evidence from real logs

The simulation controls everything; real logs control nothing, which is the point. This section replays the public Taobao display-advertising dataset [2, 3]: 26.6 million impressions over 8 days in May 2017, roughly 1.1 million users with declared demographics, and explicit campaign identifiers, the closest public analog to an operator's data. The market is reconstructed from the log itself: campaigns are the top 100 by planning-window volume, each campaign's quota is the number of impressions it actually received in the serving window, and policies re-pair the identical impression stream, so the counterfactual is *the same inventory, re-allocated*.

8.1 Honest counterfactual scoring

Replay evaluation has a trap. If the table of click probabilities that the planner optimizes also scores the outcome, both the plan and the bound harvest the table's estimation noise: in our first run the measured lift was +8.9% while the mean assignment value exceeded the best campaign's true click rate, an impossibility. The design that survives this is **dual-model scoring**: policies plan and serve using only planning-window estimates, while all values, including the hindsight bound, are scored under an independent table fitted on the serving window. A uniform-probability control run then yields exactly 0.0% lift with every policy at exactly 100% of the bound, confirming the harness measures pairing quality and nothing else. Evaluation is **walk-forward**: for each of five serving days, the plan is fitted on all prior days and scored on that day alone. The adaptive baseline is fully specified: a dual-price pacing policy that serves the quota-remaining campaign maximizing $\hat{p}_{cS} - \mu_c$, updating $\mu_c \leftarrow \mu_c + \eta(\text{served}_c/Q_c - t/T)$ after each impression with η set to half the global click rate; it consults no plan. Reported errors are the scoring table's closed-form binomial variance propagated through each comparison's assignment difference; the across-fold spread of the five per-fold lifts independently reproduces the same standard error, and the all-folds-positive sign pattern supports the parametric statement distribution-free. The uniform control defends the harness; a further check defends the scorer itself: standardizing the

scoring table along the strongest observable within-cell covariate the segmentation does not use (each user’s own training-window click propensity) shifts the headline lifts by at most +0.3pp, upward, so observable within-cell selection in the logged exposures does not account for the measured gains. The replay isolates the architecture’s allocation core, plan against serving policy at segment granularity; the per-subscriber serving components of Section 6 (soft-membership blending, frequency caps, threshold pacing) operate below this granularity and are not separately scored here.

8.2 Segment where the interaction persists

Allocation under fixed quotas can exploit exactly one thing: campaign-by-segment *interaction* in click probability that persists from the planning window to the serving window. Campaign main effects persist strongly (week-over-week correlation 0.81) but fixed quotas neutralize them; segment main effects are fixed by supply. Measuring the persistence of double-centered interaction residuals per candidate axis therefore predicts, in advance, which segmentations can pay:

Segmentation axis	Interaction persistence (r)	Interaction size beyond noise
Activity tier (usage volume)	0.48–0.54	~1pp on a 5.5pp base
Gender × age	0.28	~0.5pp
Hour × position slot	0.08	~0

The persistence table fixes the evaluation configuration *before* any replay is scored: the activity axis persists most strongly, and its measured persistence improves from four tiers to eight (r = 0.48 to 0.54), so the pre-committed configuration for the 100-campaign market is eight activity tiers. Both headline comparisons are reported for that single configuration; the remaining rows are the ablation that tests the rule.

Segments	Count	Planned vs greedy	z	Planned vs adaptive pacing	z
Activity, 8 tiers (pre-committed)	8	+3.46% ± 0.69pp	+5.0	+0.92% ± 0.47pp	+2.0
Activity, 4 tiers	4	+2.51% ± 0.50pp	+5.0	+1.15% ± 0.31pp	+3.7

Segments	Count	Planned vs greedy	z	Planned vs adaptive pacing	z
Tier × demographics	168	+1.93% ± 0.91pp	+2.1	+0.40%	+0.6
Demographics only	48	-0.20% ± 0.61pp	-0.3	+0.46%	+1.1

Three conclusions. First, the architecture’s claim holds on real inventory: on the pre-committed configuration the plan beats greedy serving by 3.5% expected clicks (positive in every fold) and beats an adaptive dual-price pacing policy in the family of online allocation methods [5], the strong online baseline, by 0.9% ($z = 2.0$; the four-tier ablation row separates from pacing at $z = 3.7$), while serving from precomputed tables. Second, **persistence, not granularity, carries the value**: eight segments on the persistent axis beat 168 finer ones, because finer cells add estimation noise and re-pairing churn faster than they add exploitable signal; and the right granularity scales with per-cell data density (in a 500-campaign market, four tiers beat eight). Third, the persistence measurement itself is the deployable design rule: before segmenting, correlate double-centered interaction residuals across adjacent planning windows, and segment only on axes that persist. The rule generalizes: on a second public log (Avazu, 40.4M mobile impressions [4]), the same measurement ranks page category as the most persistent axis ($r = 0.88$) and, in both datasets, time-of-day as the least, confirming that time belongs to the pacing loop, not to segmentation. With that rule applied, the replayed policies capture 85–91% of the hindsight bound; the residual is the estimation noise and week-over-week drift that the pacing loop of Section 6 exists to absorb.

8.3 Stress tests and the strongest baseline

Two follow-up experiments probe the market reconstruction itself, both on the pre-committed configuration. First, **over-subscription stress**: quotas set to each campaign’s realized delivery make the market exactly feasible, so we scale all quotas 25% and 50% above deliverable supply, forcing genuine contention (aggregate fill drops to 80% and 67%, and policies choose which campaigns under-deliver). The plan’s edge over greedy narrows as quotas slacken, +1.6% ($z = 2.2$) at 125% and +0.3% at 150%, which is the mechanical consequence of contention: with slack quotas, per-event value-ordering approaches per-segment optimality. The adaptive pacing policy moves the other way, degrading sharply

under contention (-3.4% and -6.4% against greedy), so the plan's advantage over it multiplies to **+5.1% ($z = 6.4$) and +7.1% ($z = 8.0$)**. The over-subscribed regime that motivates the paper is exactly where compiled planning is most valuable relative to online adaptation.

Second, the **strongest competitor**: a periodically re-solved plan, which re-solves the same LP on remaining quotas and prorated remaining supply every 20,000 impressions and serves by drawdown. It is the best policy tested, $+0.9\%$ ($z = 2.8$) over the static plan, $+1.9\%$ over adaptive pacing, $+4.4\%$ over greedy, and it is not an external rival: periodic re-solving is the pacing loop of Section 6 implemented exactly, using the same tables and the same solver. The full ordering, re-solved plan $>$ static plan $>$ adaptive duals $>$ greedy, is the architecture's recommendation made quantitative: compile the plan, serve from tables, and refresh the plan on the pacing cadence.

9 Related work

Online matching and allocation. The theoretical core of ad allocation is online bipartite matching, from the optimal $1 - 1/e$ algorithm of Karp, Vazirani, and Vazirani [6] through its budgeted AdWords generalization [7]. Under stochastic arrivals, the random-permutation model admits near-optimal algorithms that learn dual prices from an initial sample [8], and primal-dual training on forecast samples extends to display allocation [9] and general online linear programming [10]. This line of work supplies the adaptive dual-price policy we use as the strong online baseline; our contribution is orthogonal: an empirical account, on public logs, of when a compiled offline plan beats such adaptive methods and what segmentation makes the plan's advantage real.

Guaranteed display allocation. Closest in spirit are systems that precompute compact allocation plans for guaranteed-delivery display advertising: optimal online assignment with forecasts [11] and the SHALE algorithm [12] both compute offline parameters that a lightweight online server executes, as our architecture does. Those works optimize the plan's representation and solve time; we address the two questions they leave open in deployment: which audience segmentation gives the plan exploitable, durable structure (the persistence rule of Section 8), and how to measure a plan's realized value against strong baselines without an online experiment. Budget-pacing controllers [13, 14] are the production

embodiment of the adaptive alternative and correspond to the pacing loop our serving stage carries in Section 6.

Click prediction and offline evaluation. Industrial click-through-rate modeling ranges from large-scale logistic regression [15] and field-aware factorization machines [16] to attention architectures over user behavior [17, 3]. Our click model is deliberately simpler, hierarchical empirical rates with soft cluster membership, because the architecture consumes any calibrated estimator and the allocation question is about the pairing structure, not the estimator frontier. For evaluation, replay methods [18], counterfactual reasoning in computational advertising [19], and offline A/B testing [20] establish the off-policy toolbox; the dual-model scoring of Section 8.1 applies the same discipline to allocation policies specifically, where we show single-model replay inflates lift through the plan's own estimation noise.

10 Why this decomposition works

The optimization lives where time is cheap. Global allocation over all campaigns and segments is a large linear program, and it is solved where minutes are available, not milliseconds. The serving path touches five small tables and does arithmetic. This is the classic plan/execute split, applied so that the online system inherits the quality of the offline optimum at lookup cost.

Pooling defeats sparsity. No individual generates enough clicks to estimate their own response rates. Clusters do. Soft membership then re-personalizes the pooled estimates, blending each subscriber's several behavioral neighborhoods instead of forcing a single label. The same back-off logic handles new campaigns (borrow from similar campaigns) and profile-less subscribers (cluster by response history alone), so one mechanism covers three cold-start problems.

The plan is a baseline, not a cage. Real traffic never matches the forecast. By carrying planned counts into the serving state and steering with a performance index, the system tolerates forecast error gracefully: small deviations are corrected impression by impression, and persistent ones are evidence, fed back into the next modeling cycle rather than silently accumulated.

Every representation is serving-shaped. Bit masks make targeting tests branchless; pre-computed distances make personalization a weighted sum; per-(cluster, campaign) rows make pacing a two-row query. The offline stage does not merely produce a model; it compiles one, in the same sense a compiler turns a program into a form the machine executes directly.

11 Scope and assumptions

The architecture optimizes expected revenue for a known campaign portfolio over a planning horizon; it does not address auction-based pricing, real-time bidding against external demand, or creative selection within a campaign. Its guarantees rest on the stability assumptions of Section 5: allocation quality degrades in proportion to how far future traffic and response distributions drift from the modeled ones between refinement cycles, which sets the practical requirement for how frequently the offline stage must re-run. Exposure policy enforcement is per-subscriber and rule-based; richer fatigue models would slot into step 2 without changing the surrounding machinery. Finally, window-fitted planning presumes a campaign portfolio stable over the planning horizon, which is the operator's contracted-campaign setting; in marketplaces where the ad portfolio and its delivery mix turn over within a day, the re-planning interval must shrink to the churn timescale for the plan to retain value.

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